COURSE OUTLINE

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Bill Creation Automation with Claude

From a placed candidate's confirmation mail to an approved GST or non-GST vendor bill

3 days (8 hours per day) | Intermediate | Instructor-led, hands-on labs | Capstone project

About this course

Who should attend

Enterprise IT and network operations teams, finance operations and shared-services engineers, automation and platform teams in India and worldwide.

Prerequisites

- Basic Python and JSON
- Working knowledge of GST invoicing: GSTIN, CGST, SGST, IGST, HSN and SAC codes
- Familiarity with a corporate mailbox (Microsoft 365 or IMAP) and a vendor master
- Familiarity with an AI coding assistant is helpful, not required

What you will be able to do

- Describe the vendor bill workflow end to end and separate what Claude drafts from what deterministic code computes
- Set up Claude Code, the Claude Agent SDK and an MCP server that exposes mailbox and vendor-master tools with read-only defaults
- Extract candidate, vendor, amount and GSTIN fields from mail into strictly validated structured JSON outputs
- Reconcile claimed amounts against a vendor rate card and compute CGST, SGST and IGST in tested Python, never in the model
- Draft paired with-GST and without-GST previews, issue only the document that applies, gate issuance behind human approval, and keep a complete audit trail

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Tools and technologies

Claude | Claude Agent SDK | Python | Pydantic | SQLite | FastAPI | pytest | GitHub | GitHub Actions | ServiceNow | GST bill templates | Structured JSON outputs

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MODULE 1

The vendor bill workflow and the agentic baseline

- How a placed candidate's confirmation mail becomes a vendor bill today: intake, amount check, GST decision, numbering, dispatch
- What a compliant tax invoice must carry: GSTIN of supplier and recipient, place of supply, HSN or SAC, rate split, invoice number and date
- When the company issues a tax invoice with GST and when a bill of supply without it: exempt or nil-rated supplies, or when the issuing entity is not registered or under composition
- The teaching pattern: Claude plans, drafts and explains; deterministic code computes and issues; a person approves; everything is logged

HANDS-ON LAB
Walk a sample mail thread from the synthetic mailbox through the manual process and record every decision point and failure mode.

DELIVERABLE
Annotated workflow map and bill field checklist.

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MODULE 2

Setting up Claude Code, the Agent SDK and MCP for billing

- Installing Claude Code and scaffolding the billing repository with CLAUDE.md conventions for this workflow
- A Python agent on the Claude Agent SDK with tool use and a JSON schema for every response
- An MCP server exposing read-only mailbox search, vendor-master lookup and rate-card fetch as tools; no send or write tools yet
- Secrets handling: API keys and mailbox credentials in environment variables, never in prompts or the repository

HANDS-ON LAB
Build and run the MCP server, connect it to Claude Code, and have the agent list the last ten candidate mails without altering anything.

DELIVERABLE
Working Claude Code and Agent SDK environment with a read-only billing MCP server.

MODULE 3

Mail intake and structured extraction

- Reading mail through the Microsoft Graph API or IMAP, including PDF and image attachments passed to Claude for extraction
- A typed BillRequest record: candidate, vendor, amount, currency, GSTIN, service period, source message id, confidence per field
- Structured JSON outputs validated by Pydantic; unknown or conflicting fields are marked, never guessed
- Rejected-mail reporting and a queue for human clarification

HANDS-ON LAB
Run twenty synthetic mails through the extractor, including two with missing GSTINs and one with two amounts, and inspect what is flagged.

DELIVERABLE
Validated intake pipeline with a rejection report.

DAY 2 3 modules

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MODULE 4

Vendor master, amount verification and reconciliation

- Vendor master and rate card in SQLite: GSTIN, state code, registration status, agreed rates and payment terms
- GSTIN format and check-digit validation, state code derivation and vendor-status lookup in deterministic Python
- Reconciling the claimed amount against the rate card with a configurable tolerance and explicit mismatch reasons
- Claude explains each mismatch in plain language for the reviewer; it does not change the number

HANDS-ON LAB
Load the fictional vendor master, reconcile the extracted amounts, and trace one over-claim to its rate-card line.

DELIVERABLE
Reconciliation module with mismatch reasons and tests.

MODULE 5

The GST computation engine

- Intra-state versus inter-state supply: CGST plus SGST, or IGST, derived from supplier and place-of-supply state codes
- Rate slabs, HSN and SAC mapping, rounding rules and reverse-charge handling as pure functions
- The without-GST path: bill of supply for exempt or nil-rated supplies, or when the issuing entity is not registered or under composition, with the reason recorded
- Property-based and table-driven pytest coverage across every rate and state pair

HANDS-ON LAB
Implement the engine with Claude Code, review every diff, and make the test suite prove intra-state, inter-state and exempt cases.

DELIVERABLE
Tested GST engine with a full tax breakdown object.

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MODULE 6

Bill generation, numbering and templates

- Two versioned templates, tax invoice and bill of supply, rendered to PDF by deterministic code from the same BillRequest
- A FastAPI billing service with endpoints to draft, preview and (after approval) issue the one bill that applies
- Sequential, gap-free invoice numbering held in SQLite and advanced only on the approved path
- Claude drafts the covering mail and a one-paragraph reviewer summary; the numbers come from the engine

HANDS-ON LAB
Generate the with-GST and without-GST previews for the same request and diff them field by field.

DELIVERABLE
Billing service producing paired draft previews and reviewer summaries.

DAY 3 3 modules

MODULE 7

Guardrails, approvals and audit for bill issuance

- Approval package: source mail, extracted fields, reconciliation result, tax breakdown, both previews, side by side
- Human approval through a ServiceNow requested item created with the Service Catalog API, or a review CLI for teams without ServiceNow
- Hard limits: amount thresholds, duplicate-bill detection, no issuance without an approver identity
- Append-only audit log of prompts, tool calls, approvals and issued documents; retention and reviewer access

HANDS-ON LAB
Submit a bill above threshold, see it blocked, approve it as a reviewer, and read the full audit trail for that bill.

DELIVERABLE
Gated issuance workflow with audit log.

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MODULE 8

Testing, CI and accounting hand-off

- pytest suites for extraction fixtures, reconciliation and the GST engine; golden PDFs for template regression
- GitHub pull requests with Claude Code drafting changes and GitHub Actions running the suite on every pull request and push
- Exporting approved bills as CSV or Tally XML for posting, and dispatching the mail through the Graph API only after approval
- Operator runbook: re-running a mail, voiding a bill, rotating credentials

HANDS-ON LAB
Open a pull request that changes a rate slab, watch Actions fail the golden tests, fix it and merge.

DELIVERABLE
CI pipeline, accounting export and operator runbook.

MODULE 9

Capstone: end-to-end bill creation pipeline

- Wire intake, reconciliation, GST engine, templates and approval gate into one runnable pipeline on the synthetic mailbox
- Mail in, both previews drafted, the one bill that applies issued, with no issuance without approval
- Audit trail walk-through: every prompt, tool call and approval for one issued bill
- Acceptance criteria review, limitations statement, team demo

HANDS-ON LAB
Capstone build and demo.

DELIVERABLE
End-to-end pipeline and runbook.

Capstone

Capstone project

A repeatable pipeline that reads a placed candidate's confirmation mail, extracts and validates the bill request, reconciles the amount against the vendor master, computes GST in tested code, drafts both a with-GST and a without-GST preview, and issues only the one document that applies, only after a person approves, with every step logged.

Assessment

Assessment area	Weight	Evidence
Working functionality	25%	Demonstrable end-to-end workflow in a repeatable lab
Security and safety controls	20%	Threat model, authorization boundaries, approval gates, safe failure
Testing and evaluation	20%	Automated tests, golden dataset, measured quality and known limitations
Engineering quality	15%	Clear interfaces, error handling, reproducible setup, maintainable code
Evidence and explainability	10%	Traceable findings, sources, assumptions, logs
Documentation and presentation	10%	Architecture diagram, README, runbook, demo, lessons learned

Engineering principles used throughout

- Model output is a proposal, not a fact: every finding and generated change is validated by deterministic checks and tests.
- Authorization lives outside the model: tool access is enforced by application code, scoped credentials and policy.
- Human approval for consequential actions: deploying a mitigation, changing production configuration, rotating credentials.
- Decisions, evidence, model and tool versions, approvals and outcomes are logged without recording secrets.
- A model adapter and a tool adapter layer keep every build provider-agnostic and enterprise-portable.
- Quality is measured: false positives, false negatives, reliability, latency, cost and safe failure, not just a demo.

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